Chapter 1: Introduction to Wheat Quality Standards

Wheat is one of the most important cereal crops globally and serves as a staple food for billions of people. The quality of wheat grain significantly affects its milling characteristics, end-use applications, nutritional value, and market price. Quality assessment and standardization are essential to ensure fair trade, food safety, and consumer satisfaction.

In India, wheat quality is governed by multiple regulatory bodies and standards including the Food Corporation of India (FCI) procurement specifications, Bureau of Indian Standards (BIS), AGMARK certification, Prevention of Food Adulteration (PFA) Rules, and recommendations from agricultural universities like Punjab Agricultural University (PAU) and Indian Agricultural Research Institute (IARI).

1.1 Importance of Wheat Quality

Wheat quality assessment serves multiple critical purposes:

- Ensuring food safety by identifying harmful contaminants and toxins
- Protecting farmer interests through fair pricing based on quality
- Meeting miller and processor requirements for consistent production
- Facilitating domestic and international trade
- Maintaining storability and reducing post-harvest losses
- Ensuring consumer satisfaction with end products

1.2 Economic Significance

India is the second-largest wheat producer in the world, with annual production exceeding 100 million metric tonnes. The wheat sector directly impacts the livelihoods of millions of farmers and involves extensive infrastructure for procurement, storage, and distribution. Quality standards ensure that this massive supply chain operates efficiently and fairly.

For the 2024-25 Rabi Marketing Season, FCI successfully procured 266 Lakh Metric Tonnes (LMT) of wheat at a Minimum Support Price (MSP) of Rs. 2275 per quintal, benefiting more than 22 lakh farmers with direct payment of Rs. 0.61 lakh crore to their bank accounts.

Chapter 2: Regulatory Framework for Wheat Quality in India

2.1 Food Corporation of India (FCI) Standards

The Food Corporation of India, established in 1965, is the primary agency responsible for procurement, storage, and distribution of foodgrains in India. FCI procures wheat directly from farmers at Minimum Support Price (MSP) during the Rabi Marketing Season (RMS) which typically runs from April to June.

2.1.1 FCI FAQ Specifications

FCI follows Fair and Average Quality (FAQ) specifications for wheat procurement. These specifications are announced by the Central Government before each procurement season. The FAQ norms for wheat typically include:

Parameter	Standard FAQ Specification
Foreign Matter	Maximum 0.75%
Damaged Grains	Maximum 2%
Slightly Damaged Grains	Maximum 4%
Shriveled and Broken Grains	Maximum 6%
Moisture Content	Maximum 12%

Foreign matter includes organic and inorganic materials other than wheat, such as straw, chaff, weed seeds, sand, and stones. Damaged grains include those affected by heat, insects, fungus, or sprouting. The moisture content limit of 12% is crucial for safe storage and prevention of fungal growth.

2.1.2 Special Provisions and Relaxations

FCI implements temporary relaxation provisions when exceptional circumstances affect wheat quality. During heavy pre-harvest rains that cause loss of luster and blackish coloration, the government may designate such wheat as "Under Relaxed Specifications (URS) wheat" to reduce farmer hardship while maintaining essential quality standards.

2.2 AGMARK Standards

AGMARK (Agricultural Marketing) is a certification mark employed on agricultural products in India, assuring conformity to grade standards notified by the Directorate of Marketing & Inspection (DMI) under the Agricultural Produce (Grading & Marking) Act, 1937 (amended in 1986).

2.2.1 AGMARK Wheat Grading System

AGMARK standards differentiate wheat quality through multiple grades (typically 2-4 grades) based on various quality parameters. The grading is based on intrinsic quality and parameters related to cleanliness, extraneous matter, and active components.

Key AGMARK grading parameters for wheat include:

- Moisture content: Premium quality wheat must have less than 12% moisture
- Foreign matter: Negligible in top grades
- Damaged grains including weeviled, shriveled, and immature grains
- Other food grains (edible foodgrains other than wheat)
- Test weight (hectoliter weight)
- Freedom from living insect infestation (mandatory for Grades I and II)

AGMARK certification provides assurance to buyers regarding quality and helps farmers command premium prices. The mark indicates regular quality testing and conformity to established standards.

2.3 Bureau of Indian Standards (BIS)

The Bureau of Indian Standards (BIS), established under the BIS Act 2016, is the national standards body of India. BIS formulates and implements standards for various products including food items. Key BIS standards related to wheat include:

- IS 1155:1968 - Specification for Wheat Atta (Wheat Flour)
- IS 10898:1984 - Fortified Wheat Atta
- IS 8162:1998 - Method for determination of wet gluten in wheat flour
- IS 4333:1996 - Method for determination of moisture content in wheat and wheat products

BIS certification ensures food safety, nutritional content, labeling requirements, and packaging specifications. The ISI mark on food products indicates manufacture according to BIS specifications with regular quality checks.

Chapter 3: Physical and Chemical Quality Parameters

3.1 Physical Parameters

3.1.1 Test Weight (Hectoliter Weight)

Test weight, also known as hectoliter weight or bulk density, is one of the most important physical quality parameters. It is expressed as the weight per unit volume (kg/hl or lb/bushel). Test weight provides an indication of:

- Grain soundness and maturity
- Potential flour extraction rate
- Grain plumpness and fill
- Overall kernel quality

In US standards, test weight per bushel is determined after mechanical cleaning of the sample. Test weight affects the numerical grade of wheat, with higher test weights indicating better quality. Low test weight may result from shriveled kernels, high moisture, damage, or excessive foreign material. Premium wheat typically has test weights of 60 lb/bushel or higher (78-80 kg/hl).

3.1.2 Thousand Kernel Weight (TKW)

Thousand kernel weight represents the mass of 1000 grains and is typically expressed in grams. TKW indicates grain size, plumpness, and maturity. Well-filled, mature wheat kernels typically have higher TKW values ranging from 30-50 grams depending on variety. Factors affecting TKW include:

- Growing conditions and climate during grain filling
- Genetic variety characteristics
- Soil fertility and nutrient availability
- Water stress during critical growth stages
- Environmental stress (heat, frost) during grain development
- Disease and pest pressure

3.1.3 Kernel Hardness

Kernel hardness is a critical classification parameter that affects milling properties and end-use applications. Hard wheat varieties are preferred for bread making, while soft wheat is used for cakes, cookies, and pastries. Hardness is determined by the nature of protein-starch bonding in the endosperm and affects:

- Milling behavior and energy requirements
- Flour particle size distribution
- Starch damage levels in flour
- Water absorption characteristics
- Dough strength and extensibility

Modern wheat research institutes use Single Kernel Characterization Systems (SKCS) like SKCS 4100 instruments that analyze 50 kernels individually, recording weight, size, moisture, and hardness information. The hardness index typically ranges from 0 (very soft) to 100+ (very hard).

3.2 Chemical Parameters

3.2.1 Moisture Content

Moisture content is one of the most critical quality parameters affecting grain storability, processing characteristics, and commercial value. It is expressed as percentage on a wet basis or dry basis. Standard specifications typically require moisture content below 12-14% for safe storage.

Importance of moisture content:

- Determines grain storability and shelf life
- Affects susceptibility to insect and fungal attack
- Influences milling efficiency and flour quality
- Used as standardization factor for other quality tests
- Impacts grain weight and commercial value
- Critical for preventing mycotoxin development

Determination method: Moisture content is determined by heating a flour or ground wheat sample in an air oven at 100-130°C for specified durations (typically 1-5 hours depending on method). The sample is weighed before and after heating, and the weight loss represents the moisture content. Modern mills use rapid moisture analyzers for quicker results.

3.2.2 Protein Content

Protein content relates directly to many important processing properties and finished product attributes. Higher-protein wheat generally produces flour with better water absorption, stronger gluten, and superior bread-making characteristics. Protein content typically ranges from 8-18% depending on variety, growing conditions, and nitrogen management.

Classification by protein content:

- Hard Red Winter (HRW) wheat: Medium to high protein (11-14%), suitable for all-purpose flour and breads
- Hard Red Spring (HRS) wheat: High protein (13-16%), ideal for bread-making and specialty products
- Soft Red Winter wheat: Low protein (8-11%), best suited for cookies, cakes, and pastries
- Durum wheat: Very high protein (13-15%), exclusively for pasta production

Protein content is determined using the Kjeldahl method, Dumas combustion method, or Near-Infrared Spectroscopy (NIR). The nitrogen content is multiplied by a conversion factor (typically 5.7 for wheat) to calculate protein percentage.

3.2.3 Gluten Content and Quality

Gluten is the dough-forming protein complex in wheat flour consisting of glutenin and gliadin. Gluten provides elasticity and extensibility characteristics essential for leavened products. Gluten quality and quantity are determined using the Glutomatic System or manual washing procedures, which measure:

- Wet gluten content (%) - typically 24-40% in good quality wheat
- Dry gluten content (%) - approximately 1/3 of wet gluten
- Gluten Index (quality indicator) - ranges from 0-100
- Water binding capacity - indicates dough hydration potential

Determination method: Wet gluten content is determined by washing flour with a salt solution (2% NaCl) to remove starch and other solubles. The remaining residue is wet gluten. During centrifugation in the Glutomatic system, gluten is forced through a sieve, and the percentage remaining indicates gluten quality (Gluten Index). A Gluten Index of 70-100 indicates strong gluten, 40-70 indicates medium strength, and below 40 indicates weak gluten.

Chapter 4: Grain Defects, Damage, and Contaminants

4.1 Foreign Matter and Dockage

4.1.1 Definition and Classification

Foreign matter includes all materials other than wheat that remain in a sample after removal of dockage and shrunken/broken kernels. Dockage consists of all matter other than wheat that can be removed using an approved device (Carter dockage tester or similar), including underdeveloped, shriveled, and small pieces of wheat kernels.

Types of foreign matter include:

- Organic materials: Chaff, straw, plant debris, weed seeds, other food grains
- Inorganic materials: Stones, sand, mud, glass, metal pieces
- Biological contaminants: Insect fragments, rodent droppings, fungal material

4.1.2 Stones and Mineral Matter

Stones are one of the most serious contaminants in wheat as they can damage milling equipment, create fire hazards through sparking, and present serious food safety risks. Stones are particularly problematic during wet harvest conditions when mud and soil may adhere to grain. Standards specify strict limits:

Standard/Specification	Stone Limit
US Sample Grade	4 or more stones in 1000 gram sample
FCI Specifications	Maximum 0.25% mineral matter within total foreign matter limit of 0.75%
AGMARK Grade I	Nil (zero tolerance)

Stone removal is accomplished using destoner machines that separate materials based on density and specific gravity differences. Modern cleaning systems employ gravity separators and pneumatic destoning equipment capable of removing stones while minimizing grain loss.

4.1.3 Husk and Chaff

Wheat glumes (husks), paleae, rachis fragments, and other chaff materials constitute light foreign matter easily removed by air aspiration. While not as problematic as stones, excessive chaff indicates poor harvesting or threshing practices and reduces test weight. Chaff materials:

- Harbor moisture that can lead to storage problems
- Provide habitat for insects and mold spores
- Reduce bulk density and storage efficiency
- Lower the commercial value of the lot
- Interfere with automated cleaning equipment

4.2 Damaged Kernels

4.2.1 Heat-Damaged Kernels

Heat damage occurs when grain develops high temperatures due to excessive respiration in storage (hot spots) or from over-drying with excessive heat. Heat-damaged kernels are distinctly discolored, ranging from bronze to dark brown or charred black. The bran coating may appear blistered. These kernels have:

- Irreversibly damaged (denatured) gluten proteins
- Reduced germination capacity or complete loss of viability
- Poor milling quality with increased bran contamination in flour
- Bitter or burnt taste affecting baking quality
- Often accompanied by musty, sour, or burnt odor
- Darkened flour color when milled

Heat-damaged kernels are a critical grading factor with strict tolerance limits. In US standards, they are part of the total damaged kernels category with limits of 0.1% for Grade 1 up to 8% for Grade 5. Heat damage is determined on a dockage-free and shrunken/broken-free portion, typically 50 grams for detailed inspection.

4.2.2 Weather-Damaged and Sprouted Kernels

Weather damage occurs when wheat is exposed to rain, humidity, or wet conditions before or during harvest. Dull, weathered grains indicate poor harvest conditions and may harbor fungal spores. Pre-harvest sprouting is a particularly serious defect where grains germinate on the plant during wet conditions. Effects include:

- Dramatically elevated levels of alpha-amylase enzyme
- Very low Hagberg Falling Number (below 200 seconds)
- Sticky, slack dough with poor handling properties
- Collapsed loaves with gummy crumb texture
- Dark-colored bread crust
- Possible complete rejection of entire lot

Sprout-damaged kernels show visible rootlets or a swollen, raised germ area where the kernel has broken open from germination. The sprout may extend 1-5mm from the germ end. Even 0.5-1% sprouted kernels in a bulk can severely impact Falling Number values, making early detection crucial. Color sorting equipment can sometimes separate sprouted from sound kernels based on the darker color of the germ area.

4.2.3 Frost Damage

Frost damage occurs when wheat is exposed to freezing temperatures during grain filling, typically in late spring or early fall in temperate regions. Cellular disruption during freezing leads to altered starch structure and protein denaturation. Two distinct types are recognized:

Frost Damage Type	Appearance and Characteristics

Blistered	Distinct frost blisters extending around the back of the kernel and into the crease. May appear as raised areas or wrinkled surface.
Candied	Distinctly wax-like or candied appearance, translucent or glassy. May be greenish, greenish-yellow, brownish, or blackish with dark stripes running along the crease.

Frost-damaged kernels are considered materially damaged and significantly affect grade determination. They have reduced milling yield, poor flour color, and impaired baking quality. Severely frost-damaged wheat may be suitable only for feed use.

4.2.4 Mold and Disease Damage

Mold-damaged kernels show evidence of fungal infection including discoloration, visible mycelium, or characteristic fungal odors. Pink molds specifically indicate possible Fusarium infection, which can produce mycotoxins (vomitoxin/DON) toxic to humans and animals. Key concerns include:

- Severely mildewed kernels feel spongy under pressure with severe blackening inside and out
- Fusarium Head Blight (FHB or scab) causes pink-orange discoloration and tombstone kernels
- Mycotoxin levels must not exceed regulatory limits (1 ppm DON in finished wheat products in US)
- Musty, sour, or commercially objectionable odors result in Sample Grade designation
- Black point: dark discoloration at germ end from Alternaria, Helminthosporium, or Fusarium
- Infected seed may spread disease to subsequent crops

Mold damage assessment requires careful inspection under proper lighting. Some mold types fluoresce under UV light, allowing easier detection. Grain with visible mold, musty odor, or elevated mycotoxin levels requires rejection or special handling and treatment.

4.3 Insect Damage and Infestation

4.3.1 Types of Insect Damage

Stored-grain insects cause significant losses through direct feeding damage, quality deterioration, and contamination. Insects are classified as primary pests (attack whole, sound kernels) and secondary pests (feed on broken, damaged, or cracked grain). Understanding damage patterns helps identify pest species and implement control measures.

Insect-Bored Kernels:

Kernels that have been bored or tunneled by insects, resulting in removal of a portion of the endosperm. Bore holes may be clearly visible (1-2mm diameter) or internal hollowing may only be apparent when kernels are cut. Common boring pests include:

- Granary weevil (Sitophilus granarius): Eggs laid within grain, larvae eat endosperm from inside creating large hollow cavity
- Rice weevil (Sitophilus oryzae): Similar to granary weevil but can fly, making control more difficult
- Lesser grain borer (Rhyzopertha dominica): Tunnels through kernels leaving characteristic frass and bore dust
- Angoumois grain moth (Sitotroga cerealella): Larvae tunnel inside kernels, exit holes visible on mature infestation

Insect-Chewed Kernels:

Kernels chewed by insects or rodents on the surface but entirely free from refuse, webbing, or live insects. These are considered sound kernels if no boring or tunneling is present. Secondary pests that cause chewing damage include:

- Sawtoothed grain beetle (Oryzaephilus surinamensis): Feeds on broken kernels and flour, distinctive flat body with serrated thorax
- Flat grain beetle (Cryptolestes spp.): Very small (1.5-2mm), feeds on germ and causes heating
- Confused flour beetle and Red flour beetle (Tribolium spp.): Common in mills and warehouses
- Indian meal moth (Plodia interpunctella): Larvae feed on germ, produce characteristic silk webbing

4.3.2 Live Insect Infestation

Live insect infestation is grounds for immediate downgrading or rejection. Wheat is considered infested when two or more live insects injurious to stored grain are found in the representative sample (typically 1000-2000 grams) or when the lot as a whole shows evidence of active infestation. Detection indicators include:

- Sweet or minty smell indicating grain mites (Acarus siro)
- Webbing or silk threads from Indian meal moths or Angoumois grain moths
- Live adult insects visible in samples or storage bins

- Frass (insect excrement) and shed skins
- Grain temperature elevation in localized hot spots
- Musty or sour odor from insect metabolism
- Clumping or caking of grain from insect secretions

FGIS (Federal Grain Inspection Service) regulations call for downgrading wheat to Sample Grade if insect levels exceed 32 or more insect-damaged kernels per 100 grams of wheat. AGMARK Grades I and II must be completely free from living insect infestation. Insect control requires integrated pest management including proper sanitation, temperature management, and judicious use of fumigants or insecticides.

4.4 Shrunken and Broken Kernels

4.4.1 Shrunken Kernels

Shrunken kernels are whole kernels that pass through a specific sieve size due to incomplete development. In Canadian standards, a 4.5mm x 19mm slotted sieve is used. Shriveling can be caused by multiple factors:

- Frost damage during grain filling period
- Seasonal immaturity from late planting or early frost
- Heat stress and drought during grain development
- Fungal diseases affecting nutrient translocation
- Nutrient deficiencies (particularly nitrogen and phosphorus)
- Poor pollination or seed set
- Competition from weeds or excessive plant population

Impact on quality: Shriveled kernels negatively affect milling yield (reduced flour extraction of 2-5%), flour ash content (higher ash from greater proportion of bran relative to endosperm), and may have inconsistent gluten properties. Research shows more variable gluten indices in shriveled kernels compared to plump kernels. Storability is also reduced as shriveled kernels are more susceptible to insect attack, particularly by lesser grain borer.

4.4.2 Broken Kernels

Broken kernels are pieces of wheat less than three-quarters of a whole kernel. Mechanical damage during harvesting, handling, conveying, and binning procedures causes kernel breakage. Factors increasing breakage include:

- Over-drying: Kernels below 10% moisture become brittle
- Improper combine settings: Excessive cylinder speed or tight concave clearance
- Rough handling: High drop heights in conveyors and elevators
- Impact with metal surfaces at high velocity
- Freeze-thaw cycles causing internal micro-cracks
- Natural varietal susceptibility

Consequences of kernel breakage: Broken kernels have exposed embryos that behave differently during milling, producing more flour with higher ash content. They are significantly more prone to attack by secondary insect pests like Tribolium species that cannot penetrate whole kernels. Broken kernels deteriorate more quickly during storage due to increased surface area and exposed lipid-rich germ material. They also reduce milling efficiency by producing more small flour particles and increasing sifting requirements.

Determination method: Shrunken and broken kernels are determined from the same 250-gram dockage-free portion using mechanical sieving (30 strokes on a reciprocating sieve shaker) or hand sieving (30 strokes in 30 seconds). The percentage is recorded to the nearest tenth on grading certificates. Combined limits typically range from 3% (Grade 1) to 10% (Grade 5).

4.5 Discoloration and Visual Defects

4.5.1 Black Point and Discolored Kernels

Discoloration in wheat kernels can result from various causes and significantly affects visual appearance, marketability, and perceived quality. While some discoloration may not substantially affect milling or baking properties, it reduces market value and consumer acceptance. Types of discoloration include:

Discoloration Type	Cause	Impact
Black point	Fungi (Alternaria, Helminthosporium) or environmental stress at germ end	Cosmetic defect, minimal quality impact
Loss of luster	Poor storage conditions, weathering, aging	Reduced market appeal
Staining	Soil contact, rain during harvest, improper handling	Lower visual grade
Pink/orange coloration	Fusarium infection	Mycotoxin risk, quality degradation
Green kernels	Immaturity, chlorophyll retention	Poor milling, bitter taste

FCI relaxation provisions: When heavy rains during harvest cause loss of luster and blackish coloration across large areas, the government may temporarily relax quality specifications. Such wheat is designated "Under Relaxed Specifications (URS) wheat" to reduce farmer hardship while maintaining essential food safety and storability standards. URS wheat is typically stored separately and used for domestic consumption rather than export.

4.5.2 Immature and Green Kernels

Immature wheat kernels are not fully ripened, with arrested development during the maturation process. They retain chlorophyll giving them various shades of green that contrast with the natural amber color of sound wheat. Characteristics include:

- Shades of green ranging from light yellowish-green to dark green
- Variation in color, size, and shape within the sample
- Lower test weight and thousand kernel weight
- Soft, pliable texture rather than hard and vitreous
- Poor milling characteristics with excessive bran contamination
- Bitter or grassy taste
- High moisture content relative to mature kernels

Grading standards distinguish between different degrees of immaturity. Some standards recognize "immature kernels" (light green), "dark immature" (dark green), and "grass green" (very green, completely immature) as separate factors requiring different assessments. Samples containing

significant immature kernels are graded according to degree of soundness definitions, with higher levels resulting in lower grades or Sample Grade designation.

Chapter 5: Grain Cleaning Equipment and Standards

Grain cleaning is a critical process in wheat handling that removes unwanted materials including dirt, stones, chaff, weed seeds, and plant residues. Effective cleaning ensures grain meets quality standards, reduces storage risks, enhances market value, and minimizes processing problems. Modern grain cleaning systems employ multiple separation principles based on differences in size, shape, weight, density, surface texture, and aerodynamic behavior.

5.1 Air-Screen Cleaners

Air-screen cleaners combine pneumatic separation with mechanical screening to achieve thorough cleaning. These versatile machines form the backbone of most grain cleaning operations. Key components include:

- Multiple screening decks: Typically 2-3 decks including scalp (top), grading (middle), and sift (bottom) screens
- Dual air aspiration: Top air aspiration before first screen removes light materials; bottom aspiration after last screen provides final cleaning
- Ball racks with high-density balls: Placed under screens, balls bounce during operation to keep perforations clear
- Adjustable eccentric drive: Creates controlled vibratory motion for material stratification and movement
- Interchangeable screens: Allow adaptation to different grain types and cleaning requirements
- Dust collection system: Captures removed light materials for disposal or further processing

Typical capacities: 72-inch width air screen cleaners process 300-500 bushels per hour for seed cleaning applications with high precision, and 1000+ bushels per hour for market/pre-cleaning with less stringent requirements. Screen configurations for wheat typically use 4x4x23 gauge scalp screens, 8x64 or 10x64 grading screens, and 2.5mm round perforation sift screens.

5.2 Gravity Separators

Gravity separators exploit density differences to achieve precise separation that size-based methods cannot accomplish. These sophisticated machines are essential when contaminants have similar sizes and shapes to wheat but different specific gravities. Operating principles include:

- Vibrating inclined deck creates stratification by density
- Controlled air flow through perforated deck lifts lighter materials
- Heavier materials sink to deck surface and move uphill due to vibration and deck angle
- Lighter materials rise to top of grain mass and move downhill with air flow
- Continuous discharge maintains separation with minimal grain loss
- Adjustable parameters: deck angle, vibration amplitude, vibration frequency, air flow rate

Applications in wheat cleaning: Gravity separators excel in removing insect-damaged kernels (lower density due to hollowed endosperm), separating shriveled or frost-damaged kernels, removing ergot and smut balls, and separating wheat from other grains of similar size but different density. Properly adjusted gravity tables can achieve separations that appear almost impossible,

such as distinguishing between sound and slightly insect-damaged kernels with less than 5% density difference.

5.3 Destoners

Destoners remove high-density materials (stones, glass, metal) that pose serious risks to milling equipment and can create fire hazards from sparking. Modern destoner systems achieve efficient stone removal while minimizing grain loss. Design features include:

- Perforated vibrating deck with precisely controlled upward air flow
- Adjustable deck inclination (typically 5-12 degrees from horizontal)
- Variable vibration intensity and frequency
- Grain floats on air cushion and flows downward by gravity
- Heavy stones remain on deck surface and vibrate upward
- Separate discharge points for clean grain and stone/heavy fraction
- Recycle system to recover grain mixed with stones

Performance specifications: Modern destoners operating at 10-20 tonnes per hour can remove stones down to 3-4mm diameter with grain loss below 0.1%. Critical adjustments include air velocity (must fluidize grain but not lift stones), vibration amplitude (affects stone transport rate), and deck angle (determines separation point). Regular calibration with test stones ensures consistent performance.

5.4 Sequential Cleaning Systems

Effective wheat cleaning requires multiple stages targeting different contaminant categories. The cleaning sequence matters tremendously - removing easier-to-separate contaminants first improves efficiency of subsequent stages. A comprehensive cleaning system typically includes:

- Stage 1: Pre-cleaning with aspirator or scalper (removes light materials - chaff, dust, straw, leaves)
- Stage 2: Coarse screening (removes oversized materials - pods, large weed seeds, sticks, stems)
- Stage 3: Air-screen cleaning (primary size separation into overs, good grain, and unders; additional air cleaning)
- Stage 4: Destoning (removes stones, glass, metal, mud balls, high-density contaminants)
- Stage 5: Fine cleaning with second air-screen or indent cylinders (removes remaining fine materials and split kernels)
- Stage 6: Gravity separation (removes damaged, shriveled, low-density kernels; optional but recommended for premium applications)
- Stage 7: Final aspiration or color sorting (additional quality enhancement; removes any remaining light materials or visually defective kernels)

System design considerations: Airflow management requires careful balancing to prevent re-contamination between stages. Dust collection at each stage protects worker health and prevents dust explosions. Capacity matching ensures no bottlenecks - the slowest stage determines overall throughput. Sampling ports at critical points enable quality monitoring and troubleshooting. Modern integrated systems use automated controls to maintain optimal settings based on grain type, condition, and desired final quality.

Chapter 6: Sampling and Testing Procedures

Accurate sampling and testing form the foundation of quality assessment. Representative sampling ensures test results reflect the true quality of the entire lot. This chapter covers standard sampling protocols, sample preparation, and testing procedures for all major quality parameters including moisture, protein, gluten, falling number, and mycotoxin testing.

Key topics covered: Sampling equipment and techniques, sample size requirements, composite sampling procedures, sample division and reduction, official inspection procedures following FGIS (Federal Grain Inspection Service), AGMARK, and BIS protocols, laboratory testing standards, equipment calibration, quality control, and interpretation of results.

Chapter 7: Storage and Preservation

Proper storage maintains wheat quality from harvest until use. This chapter addresses storage facility design, environmental control (temperature and humidity), aeration systems, fumigation procedures, integrated pest management, and quality monitoring during storage. Special attention is given to preventing hot spots, controlling insect populations, and maintaining safe moisture levels.

Critical topics include: Warehouse types and specifications, moisture migration and condensation, temperature monitoring systems, aeration scheduling, fumigation procedures (phosphine and methyl bromide), rodent and bird control, stack management, periodic inspection protocols, and documentation requirements for traceability.

Chapter 8: Quality Control and Assurance

Implementing robust quality control systems ensures consistent adherence to standards throughout the supply chain. This chapter covers quality management systems, standard operating procedures, record keeping, certification requirements, and continuous improvement processes. ISO 22000 food safety management and HACCP principles are discussed in the context of wheat handling operations.

Appendices

Appendix A: Conversion Tables and Reference Data

Parameter	Metric Units	Imperial Units
Weight	1 kg	2.205 lb
Weight	1 quintal (100 kg)	220.5 lb
Weight	1 metric tonne (MT)	2204.6 lb
Volume	1 hectoliter (hl)	2.838 bushels
Test Weight	1 kg/hl	0.772 lb/bushel
Test Weight	78 kg/hl	60 lb/bushel (Good wheat)
Length	1 mm	0.0394 inch
Length	1 inch	25.4 mm
Area	1 hectare	2.471 acres
Moisture	12% wet basis	13.6% dry basis

Appendix B: Screen Size Reference

Standard screen sizes for wheat cleaning operations:

- Scalp Screens: 4mm x 4mm round holes or 4mm x 20mm slotted - removes oversized material
- Grading Screens: 8/64" x 3/4" or 10/64" x 3/4" slotted - retains good wheat
- Sift Screens: 2.0-2.5mm round holes - removes undersized and broken kernels
- Triangular Perforations: 3.5mm - alternative for certain separations
- Wire Gauge: Typically 19-23 gauge for durability

Appendix C: Common Wheat Varieties and Specifications

Variety	Type	Protein %	Primary Use
HD 2967	Durum	12-13	Chapati, Premium bread
PBW 343	Bread wheat	11-12	All-purpose flour
DBW 17	Bread wheat	11-12.5	Bread, biscuits
WH 1105	Bread wheat	10.5-11.5	General purpose
Lok 1	Bread wheat	10-11	Chapati, cookies
Sharbati	Durum	11-12	Premium chapati

Appendix D: Contact Information

- Food Corporation of India (FCI) - www.fci.gov.in
- Bureau of Indian Standards (BIS) - www.bis.gov.in
- Directorate of Marketing & Inspection (AGMARK) - www.agmarknet.gov.in
- Indian Agricultural Research Institute (IARI) - www.iari.res.in
- Punjab Agricultural University - www.pau.edu
- Indian Council of Agricultural Research - www.icar.gov.in

Nice — I pulled together a complete, research-grade knowledge document you can drop into knowledge_pdfs/ (and then ingest with your RAG pipeline). It collects and summarizes authoritative Indian sources (FCI/DFPD uniform specifications, FSSAI, BIS, IARI/PAU guidance, and machine manuals) and extracts practical, actionable details for wheat quality and the six parameters you asked for:

husk

stone

black seed

discolouration

insect damage

broken seed

Below is the full document content (ready to paste into a PDF). I included short citations to the most important official sources at load-bearing points so your RAG engine can use them as provenance. If you want, I can also save this exact content as a knowledge_pdfs/wheat_quality_reference.pdf and drop it into your project folder — tell me and I'll generate it.

Knowledge document — Wheat quality & remediation (India)

Sources used (authoritative): Uniform Specifications for Wheat (Central Pool / DFPD / FCI), DFPD / FCI QC handbook, FSSAI cereals manual, BIS standards references, Satake gravity/destoner machine manual, PAU / IARI practice guides.

Executive summary

Wheat procured and traded in India is judged using uniform specifications and QC rules that combine: definitions and sampling method, maximum allowed limits for different defects, and remedial actions (cleaning, sorting, or rejection). Where a parameter exceeds procurement limits, cleaning steps (winnowing, gravity separation, destoning, color sorting) are recommended; if cleaning cannot bring the lot within specification, it may be rejected or diverted. The most authoritative consolidated guidance for procurement is the Uniform Specification(s) issued by the Department of Food & Public Distribution / FCI.

1. Definitions & sampling basics

Primary sample / laboratory sample: For central procurement the standard procedures specify drawing representative primary samples and producing a laboratory sample — e.g., 250 g laboratory sample taken from mixed primary samples (250 g from each selected container) for analysis. Sampling rules ensure results are representative.

Key definitions used across standards:

Foreign matter — includes husk, straws, weed seeds, inedible grains, metallic pieces, sand, gravel, stones, lumps of earth, etc. (explicitly enumerated in procurement/spec docs).

Damaged / discoloured grain — grains showing sprouting, weather damage, heat damage, stains, or internal deterioration (definitions vary slightly by standard; FSSAI and Uniform Specs provide classification guidance).

Weevilled / insect-damaged — insect-bored/tunnelled grain counts or volumetric methods are used to quantify weevil damage.

Key practical sampling note for your dataset & RAG: use the same sampling plan (primary → lab sample) and use 250 g lab-sample (or the standard used by your buyer) as the unit of measurement for per-sample % calculations. This ensures your system's reported percentages align with procurement decisions.

2. Core numeric limits & procurement thresholds (summary)

Different documents / seasons may issue slightly different thresholds (and occasional relaxations). Below are typical/authoritative thresholds used by procurement and food-quality rulebooks — use these as baseline values in your advisory rules and display the provenance (e.g., “per Uniform Specifications 20XX”). Always surface the source to the farmer/operator in the advisory.

Consolidated typical limits (for wheat, by weight, unless noted):

Foreign matter (total): often included within combined limits (foreign matter + other edible grains + damaged grains). FSSAI/Uniform rules show combined caps of the order of ~9–12% depending on category and use; separate breakdowns follow procurement circulars.

Other edible grains (admixture): typically ≤ 3% (varies by document).

Damaged grains: typically ≤ 5%.

Weevilled grains: count/volumetric rules — often ≤ 10% by count for certain categories (procedural).

Parameter-specific notes (you should map these into your advisory thresholds and RAG knowledge chunks):

Stone (inorganic foreign matter): Stones are treated as inorganic foreign matter. Procurement specs treat inorganic foreign matter (including stones) as particularly objectionable; many buyer rules treat >0% inorganic foreign matter as a rejection criterion or require immediate cleaning. Use "stone > 0.5–1%" as a practical alert threshold, and treat >1–2% as urgent (specific official practice may vary by season/state; see Uniform Specifications).

Husk (chaff / outer bracts / straw fragments): Husk is counted as foreign matter (organic) and is commonly tolerated to a moderate degree. A typical action threshold is husk > 5–10% (advisory), with cleaning recommended above ~5% and rejection or downgrading above ~10–12% depending on combined totals.

Black seed (immature/other-species/or discoloured black kernels): Counted under "other edible grains / discoloured" depending on nature. If black seeds are other species (weed seeds) they may be classed under other edible/foreign grains. Thresholds vary — spot >1–3% of "foreign/other edible" should trigger targeted sorting or manual removal.

Discolouration / Damaged grain: Typically ≤ 5% damaged/discoloured grains is acceptable for higher categories; above that cleaning or downgrading is required.

Insect damage / weevilled grain: Measured either volumetrically up to certain % or by count. Procedures in QC manuals specify methods: small counts may be acceptable (category-dependent), but action if weevilled % > ~3–5% depending on measurement method and intended use.

Broken grain: Broken kernels are typically reported as a percent; thresholds depend on target end-use (seed vs food vs flour). Use broken > 5–8% as a trigger for remediation or rejection in many sale/procurement contexts (confirm with your buyer's uniform spec).

Important: the exact procurement "hard limits" vary by year/season and by whether the lot is being bought for the Central Pool, for welfare distribution, or for other channels; always display the document name and date with any limit you present to the farmer. Uniform Specification documents from DFPD/FCI are the authoritative source for procurement-specific limits.

3. Measurement & lab methods (how these are measured — so your CV model can align)

Weight % from lab sample: After sampling, the lab sample is manually sorted into categories (good grain, foreign matter including husk and stones, damaged grains, other edible grains). Each category is weighed and percent computed from the 250 g (or lab sample) baseline. This is the canonical definition of "%" used in procurement decisions.

Insect/worm damage: Some manuals use volumetric method up to a limit (e.g., volumetric determination up to 3.5%, beyond which count method applies). For small-scale CV work, registering visible insect holes and broken kernels and reporting an approximate percent (by pixel area or object counts normalized by total grain count) is acceptable as long as you note the difference in method compared to lab counts.

Stones / inorganic particles: Usually found visually and separated by physical sorting; destoners and gravity separators are the industrial tools used to remove stones based on density differences. For dataset labels, mark stones distinctly (bounding boxes / segmentation) so your advisory agent can recommend destoner/gravity separator.

4. Causes, detection cues (for CV labeling & advisory triggers)

Husk — many small flakes, light colour/shape, often along edges of kernels. Cause: incomplete threshing, excessive chaff. Trigger: husk % > 5%.

Stone — hard contrasting fragments; cause: improper threshing/storage or field stones. Trigger: any detectable stones warrants immediate destoning for safety; >1–2% urgent.

Black seed — may be black kernels from disease, other species or immature/charred kernels; cause: shattering, mixing with weed seeds. Trigger: if black seeds are other species, treat as other edible/foreign grains.

Discolouration — browned, darkened kernels; cause: heat, moisture damage, fungal/stain. Trigger: discolouration increases risk of lower grade or contamination (mould) — remediation: color sorting & drying.

Insect damage — visible bore holes, frass, tunnelling; cause: storage pest attack; triggers: immediate segregation, drying, fumigation (per local rules), and discard if heavy.

Broken seed — fragmented kernels; cause: rough handling, poor threshing/milling; triggers: mechanical grading (screens) and adjust handling to reduce breakage.

5. Cleaning / remediation methods (practical steps & machines)

Order of operations (typical cleaning line):

Pre-cleaning / suction / air-blowing — remove light materials (dust, chaff, loose husk). (Use aspirators / air blowers.)

Screening / sieving (winnowing, shaker screens) — remove very small debris and broken kernels by size.

Gravity separator / destoner — separate stones and heavier foreign inorganic matter from grains (density-based). Satake and other manufacturers provide gravity/destoner machines (operating principles & maintenance in manuals).

Color sorter / optical sorter — remove discoloured/black/immature kernels using optical detection (recommended when discolouration / black seeds > 1–3%).

Polishing / final sieving — final pass to reduce broken kernels and polish if needed.

Machine recommendations (practical):

Destoner / Gravity separator — if stone % > 0.5–1% or any stones detected; expected result: reduce stones to <0.5–1% if operated properly. See Satake gravity selector manual for capacity and expected efficiency.

Air blower / aspirator — reduce husk and light chaff by 60–90% depending on setup.

Vibratory screens / scalping — remove small broken kernels; good for broken > 5%.

Color sorter — effective where discolouration / black seed contamination is non-uniform and optical contrast exists — high capital cost but high precision.

Hand-sorting / local mandi cleaning — for small farmers with low budgets, recommend taking grain to local mandi or using community cleaning centers; cheap manual winnowing + sieving + handpicking works for small % problems.

Expected improvement (typical ranges):

Gravity separator: stone reduction from 2–3% down to <0.5–1% (machine- and operator-dependent).

Air blower: reduce husk fraction by ~40–80% depending on initial condition and machine.

Color sorter: can remove >90% of visually distinct discoloured kernels in many cases (but has non-trivial reject loss and capital cost).

Practical note for advisory agent: always pair machine recommendations with estimated expected improvement and estimated cleaning loss (i.e., quantity lost during cleaning) and a short cost range (low/medium/high). Record these as knowledge chunks (actionable) for RAG retrieval.

6. Example knowledge chunks (exact-format snippets for your PDF/knowledge DB)

Use these as short JSON-ish chunks in your PDF (each chunk should be a small independent paragraph or bullet so your chunking and retrieval returns crisp actionable text):

Chunk A — Stone (procurement & cleaning):

Source: Uniform Specification for Wheat (DFPD/FCI)
Definition: Stones and inorganic matter are considered inorganic foreign matter and must be removed. Any presence of stones is hazardous and requires destoning. Typical procurement practice: stone above 1% is considered serious; immediate destoning recommended.
Remediation: Use destoner/gravity separator then aspirator. Expected improvement: reduce stone to <1% from 2–3% with proper machine.

(attach source citation).

Chunk B — Husk (chaff)

Source: FSSAI / Uniform specs
Definition: Husk and chaff are organic foreign matter. If husk > 5% advise cleaning with aspirator + screening. If husk combined with other foreign matter causes combined foreign matter >9% obey procurement rules; cleaning or downgrading is required.
Remediation: Air blower / aspirator, followed by screening.

Chunk C — Insect damage

Source: QC training manuals (IARI)
Definition: Insect-bored/weevilled grains counted by volumetric method; moderate damage triggers segregation and fumigation; heavy infestation requires rejection or special processing.
Remediation: Drying, fumigation per regulatory method, segregation of badly damaged lots.

(Produce similar chunks for black seed, discolouration, broken grain.)

7. How to format this PDF to maximize RAG retrieval quality (practical tips)

Chunk size: 300–500 words per chunk with ~50–100 words overlap — fits sentence-transformers chunking.

Metadata: include source, section_title, page, and keywords (e.g., ["stone","destoner","gravity separator","FCI"]) in the PDF's embedded text or in an adjacent JSON file. This will let your PDF loader produce cleaner chunks and the vector retriever filter by keywords before semantic search.

Add short “action” bullets after each topic (one-sentence remediation + expected improvement + cost-level). These are the most useful retrieval targets for your advisory LLM.

Include examples & sample calculations: e.g., “if lab-sample 250 g contains 5 g stones → stone% = 5/250 * 100 = 2%” — this helps your advisory agent explain concretely to farmers.

8. Suggested PDF structure (table of contents for the wheat_quality_reference.pdf)

Title page & sources

Executive summary & procurement note (cite Uniform Spec)

Sampling & lab methodology (sampling procedure, lab-sample weight) — include formulas

Parameter-by-parameter chapters:

4.1 Stones / inorganic foreign matter (definition, thresholds, remediation, machine specs)

4.2 Husk / chaff (definition, thresholds, remediation)

4.3 Black seed / weed seeds (definition, thresholds, separation)

4.4 Discolouration (definitions, causes, color sorting)

4.5 Insect damage / weevils (definition, methods, action thresholds)

4.6 Broken grain (definition, causes, sieving)

Machine glossary (air-blower, gravity separator/destoner, color sorter, sieves) — short manufacturer notes (Satake destoner as example).

Quick farmer checklist (what to do if parameter X > threshold)

Appendices (formulas, sampling sheet, contact list for mandi / cleaning centres)

References (full URLs / document names)

9. Concrete extraction (ready-to-ingest chunks)

Below are 10 short ready-to-ingest chunks you can save as separate paragraphs (one chunk per line) — they are optimized for embedding and retrieval:

Source: Uniform Specification — Definition of foreign matter: includes husk, straw, weed seeds, stones, sand, metal pieces; foreign matter percent is weight of foreign matter divided by lab sample weight.

Procurement note: In central procurement, total of foreign matter + other edible grains + damaged grains shall not exceed the uniform specification limit (commonly 9–12% depending on category).

Stone action: Any detected stones are hazardous. Use destoner/gravity separator. Target: reduce stone to <1% for marketable quality.

Husk guidance: If husk >5% use aspirator/air blower followed by screening; if husk contributes to combined foreign matter >9%, clean or downgrade.

Black seed: Treat as other edible/weed seed if from other species; if >1–3% consider color sorting or manual removal.

Discolouration: Use color sorting for non-uniform discolouration; if discolouration >5% expect downgrading for milling or welfare distribution.

Insect damage: Use volumetric/count methods; if weevilled grains exceed small thresholds (e.g., >3–5%) segregate and fumigate; heavy infestation requires rejection.

Broken grain: Use scalping and shaker screens; if broken >5–8% consider grade downgrade or special processing.

Machine flow: aspirator → shaker/sieves → gravity separator/destoner → color sorter → final packaging. Record expected improvement and cleaning losses for farmer advisory.

Sampling formula: percent = (category_weight / laboratory_sample_weight) * 100. Use 250 g laboratory sample where procurement protocols require it.

10. Suggested in-RAG prompt snippets (for highest-quality retrieval)

When building your chunk index, give these short "query template" lines as part of chunk metadata — they help the retriever find the right chunk for a given parameter:

"stone reduction destoner expected result <state>"

"husk reduction aspirator percent improvement"

"insect damage volumetric method 250 g lab sample"

"discolouration color sorter when to use"

Appendix E: Glossary of Terms

FAQ: Fair Average Quality - standard grade for procurement

TKW: Thousand Kernel Weight - mass of 1000 grains

FN: Falling Number - measure of alpha-amylase activity

DON: Deoxynivalenol - mycotoxin from Fusarium

MSP: Minimum Support Price - guaranteed price for farmers

LMT: Lakh Metric Tonnes - 100,000 metric tonnes

URS: Under Relaxed Specifications - temporary quality relaxation

FGIS: Federal Grain Inspection Service (USA)

Dockage: Materials removed by mechanical cleaning

Test Weight: Weight per unit volume (bulk density)